

ISRO Hackathon

Air-Gapped Predictive Copilot for Secure MPLS Operations

Master Technical Vision, Research Blueprint & Implementation Strategy

1. Vision Statement

Modern Network Operations Centers (NOCs) remain fundamentally reactive.

While current monitoring platforms can identify faults after they occur, operators still spend significant time correlating alerts, identifying root causes, estimating impact, and selecting remediation actions.

The objective of Project AEGIS is to transform NOC operations from a reactive monitoring paradigm into a predictive decision-support paradigm.

The system shall continuously analyze network telemetry, routing behavior, SD-WAN tunnel health, topology structure, and historical operational patterns to:

1. Predict failures before service degradation occurs.
2. Estimate time-to-impact before SLA breach.
3. Explain the reasoning behind predictions.
4. Prioritize operational risks.
5. Recommend corrective actions.
6. Operate entirely within an air-gapped environment.

The final platform acts as an AI-assisted Network Operations Engineer rather than a traditional monitoring dashboard.

2. Strategic Objectives

The platform must continuously answer:

Q1. What is likely to fail next?

Examples:

- MPLS congestion
- Route instability
- Tunnel degradation
- Policy drift
- Device failure

Q2. Why is the risk elevated?

The system must identify:

- Contributing telemetry signals
- Routing anomalies
- Topology vulnerabilities
- Historical pattern matches

Q3. What should operators do before impact occurs?

The platform must provide:

- Recommended actions
- Ordered execution sequence
- Expected impact reduction
- Confidence level

3. Design Philosophy

The architecture follows six principles:

Principle 1 — Predict Before Detect

The objective is not alert generation.

The objective is degradation forecasting.

Principle 2 — Human-Centered Operations

AI recommends.

Operators decide.

The platform remains advisory rather than autonomous.

Principle 3 — Explainability First

Every prediction must be explainable.

Operators should never receive:

"Risk Score = 92%"

without supporting evidence.

Principle 4 — Offline by Design

No cloud APIs.

No external inference services.

No outbound connectivity.

All intelligence remains within the air-gapped boundary.

Principle 5 — Multi-Layer Intelligence

No single model can understand a network.

Forecasting, topology analysis, routing behavior, and operational knowledge must be combined.

Principle 6 — Operational Relevance

Outputs must resemble real NOC incident reports rather than ML predictions.

4. System Architecture

Network Infrastructure



Telemetry Collection Layer



Data Processing Layer



Forecasting Layer



Graph Intelligence Layer



Risk Fusion Layer
↓
Explainability Layer
↓
Knowledge Retrieval Layer
↓
Offline Copilot Layer
↓
Operator Dashboard

5. Network Environment

The simulated environment should represent a realistic enterprise WAN.

Components

Branch Sites

Hub Sites

Datacenters

Controllers

Core MPLS Infrastructure

Device Roles

CE Routers

PE Routers

P Routers

SD-WAN Controllers

Edge Devices

Network Services

MPLS Forwarding

VPN Segmentation

Traffic Engineering

Dynamic Routing

IPSec Overlay Tunnels

QoS Enforcement

Application-Aware Routing

6. Telemetry Architecture

The telemetry pipeline serves as the sensory system of the platform.

Telemetry Sources

SNMP

Interface statistics

CPU

Memory

Error counters

NetFlow/IPFIX

Traffic behavior

Application flows

Bandwidth consumption

Flow duration

Syslog

Device events

Configuration changes

Routing events

Authentication events

Streaming Telemetry

SD-WAN Controller Statistics

Tunnel Metrics

Performance Metrics

Policy Updates

7. Dataset Design

The project should generate a structured dataset consisting of five analytical layers.

Layer 1 — Infrastructure Metrics

- CPU Utilization
 - Memory Utilization
 - Interface Utilization
 - Error Counters
 - Queue Drops
 - Buffer Usage
-

Layer 2 — Network Quality Metrics

- Latency
- Jitter
- Packet Loss
- Throughput
- Tunnel Availability
- Tunnel Rekey Events

Layer 3 — Routing Metrics

- BGP Updates
- Route Advertisements
- Route Withdrawals
- Route Flap Count
- OSPF State Changes
- Convergence Time

Layer 4 — Traffic Flow Metrics

- Active Flows
- Top Applications
- Flow Duration
- Bytes Per Flow
- Protocol Distribution

Layer 5 — Graph Features

Generated dynamically.

- Degree Centrality
- Betweenness Centrality
- Closeness Centrality
- Path Redundancy
- Edge Importance
- Node Criticality Score

8. Predictive Analytics Framework

The predictive engine consists of multiple specialized components.

Module A — Time-Series Forecasting

Purpose:

Predict future network conditions.

Prophet

Predicts:

- Latency trends
- Utilization trends
- Jitter trends

Advantages:

- Fast
 - Explainable
 - Low computational overhead
-

LSTM

Predicts:

Future network state vectors.

Inputs:

- Utilization
- Latency
- Jitter
- Packet Loss
- Routing Events

Outputs:

Future degradation probabilities.

Module B — Routing Stability Analytics

Purpose:

Identify precursor conditions of routing failures.

Features:

- Route flap frequency
- Adjacency resets
- Convergence delays

- Advertisement spikes

Output:

Routing Instability Score

Module C — Tunnel Health Analytics

Purpose:

Predict SD-WAN tunnel degradation.

Features:

- Packet loss progression
- Jitter growth
- Tunnel throughput
- Rekey anomalies

Output:

Tunnel Risk Score

9. Graph Intelligence Layer

The network is inherently a graph.

Ignoring topology removes critical context.

Graph Construction

Nodes:

Routers

Sites

Controllers

Edges:

MPLS Links

IPSec Tunnels

Routing Relationships

Graph Embeddings

Node2Vec

Graph Features

Topology Fingerprints

Graph Anomaly Detection

Isolation Forest

Output:

Topology Risk Score

10. Risk Fusion Engine

Purpose:

Combine all analytical perspectives.

Inputs:

- Prophet Outputs
- LSTM Outputs
- Routing Risk
- Tunnel Risk
- Graph Risk
- Current Telemetry

Model:

CatBoost Meta-Ensemble

Outputs:

- Risk Score
 - Confidence Score
 - Failure Class
 - Severity
-

11. Time-to-Impact Engine

Purpose:

Estimate remaining time before operational impact.

Output Examples:

Congestion:
22 Minutes

Tunnel Failure:
15 Minutes

Route Instability:
9 Minutes

This becomes one of the most valuable operational metrics.

12. Explainability Framework

Every prediction must be explainable.

Implementation:

SHAP

Feature Attribution

Decision Tracing

Example

Risk Score:
93%

Contributing Factors:

- Utilization Growth (+34%)
 - Packet Loss Trend (+22%)
 - Routing Instability (+18%)
 - Tunnel Health Decline (+11%)
-

13. Remediation Intelligence Framework

Prediction alone is insufficient.

The system must provide actionability.

Remediation Knowledge Base

Sources:

Runbooks

SOPs

Incident Reports

Vendor Documentation

Best Practices

Action Ranking Engine

For every detected issue:

Generate:

- Recommended actions
 - Execution order
 - Expected effectiveness
-

Example

Predicted Issue:
Hub Congestion

Recommended Actions:

1. Reroute backup traffic
2. Adjust QoS policies
3. Delay bulk synchronization jobs

Expected Risk Reduction:
67%

14. Offline LLM Copilot

Purpose:

Translate analytics into operator-ready intelligence.

Recommended Models

Phi-3 Mini

Mistral 7B Quantized

Responsibilities

Incident Summarization

Root Cause Narratives

Action Recommendations

Natural Language Queries

Operational Guidance

15. Retrieval-Augmented Generation

The LLM must not rely on internal weights alone.

Knowledge Sources:

- Network Topology

- SOPs
- Runbooks
- Historical Incidents
- Device Documentation

Vector Database:

FAISS

or

ChromaDB

16. Operator Dashboard

Core Panels:

Network Topology

Predicted Risks

Time-to-Impact

Tunnel Health

Routing Stability

Recommended Actions

Incident Timeline

Copilot Chat Interface

17. Fault Injection Strategy

The dataset must contain realistic operational failures.

Scenario 1

Progressive Congestion

Scenario 2

BGP Route Flapping

Scenario 3

OSPF Convergence Instability

Scenario 4

Tunnel Degradation

Scenario 5

Packet Loss Escalation

Scenario 6

Controller Policy Drift

Scenario 7

QoS Misconfiguration

Scenario 8

Multiple Simultaneous Faults

18. Evaluation Framework

Technical Merit

Prediction Accuracy

Precision

Recall

F1 Score

False Positive Rate

Prediction Lead Time

Copilot Quality

Explanation Accuracy

Grounding Score

Action Relevance

Operational Clarity

Security

Zero Outbound Connectivity

Offline Deployment Verification

Model Packaging Compliance

19. Innovation Highlights

Innovation 1

Forecasting + Graph Analytics Fusion

Innovation 2

Time-to-Impact Estimation

Innovation 3

Risk Fusion Meta-Ensemble

Innovation 4

Explainable Network Intelligence

Innovation 5

Remediation Intelligence Layer

Innovation 6

Graph-Aware Failure Prediction

Innovation 7

Fully Air-Gapped Copilot Architecture

20. Final Success Criteria

The platform shall successfully function as a predictive NOC copilot capable of:

- Forecasting failures before SLA impact.
- Explaining risk factors with transparency.
- Prioritizing operational threats.
- Providing remediation guidance.
- Reducing alert fatigue.
- Supporting operator decision-making.
- Operating entirely within an air-gapped environment.

The final outcome is not a monitoring tool.

The final outcome is an intelligent operational decision-support platform capable of augmenting human network engineers in mission-critical environments.